K56: Cathedral Component-Level Δr Table — Per-Entry Tier Review

Keeper

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Context

The Cathedral Component-Level Closure (Section 7.1 of Paper #115 v0.5, supported by Elie Toy 3012 from Monday May 18) tabulates Δr decomposition for fundamental observables, with several entries labeled D-tier at 1-2% precision.

Cal's K55 walk-back audit (Tuesday) flagged this as Tier-C: per his external-survivability checklist, D-tier labels require exhibited mechanism chains to D_IV^5, not just BST-primary-decomposability at sub-percent precision. A 1-2% match between observed quantity and a BST-primary expression is structurally suggestive but does NOT close the mechanism gap.

This is the same discipline shape as Cal's verdict #23 yesterday (muon g-2 stepped down D→I) — striking precision identification ≠ mechanism derivation.

What the Cathedral Δr table currently carries (snapshot from Section 7.1 + Toy 3012)

Lyra's read-pass on Paper #120 §4.4 yesterday already calibrated one entry (Newton's G hierarchy) to "structural identification with multi-week numerical closure open." That calibration pattern is what's needed at the COMPONENT level across all Δr entries.

Per Cal flag a, several specific identifications carry D-tier label without explicit mechanism citation:

Δr entry	Observable	BST primary form	Precision	Current label	Mechanism cited
1	α^{-1}(M_Z) − N_max	N_c^2 (BST primary)	0.5%-class	D-tier in current table	Per Cal #20 yesterday: stepped down 0.0004% → 1.4% on correction term, structural mechanism not fully derived
2	$\Delta\alpha =$ N_c^2/N_max	substrate-coupling	sub-percent	D-tier	mechanism partial (Cathedral 12/12 component-level closure framing)
3	$\Delta\rho = 1/107$	substrate residue	structural	D-tier	mechanism not exhibited (107 not BST primary; relation to BST integers needs derivation)
4	y_t = 1 − 1/N_max = 136/137	top Yukawa	<1%	D-tier	mechanism not fully exhibited (why y_t specifically vs other Yukawas?)

Δr entry	Observable	BST primary form	Precision	Current label	Mechanism cited
5	Δr_rem = -1/726	residual	structural	D-tier	mechanism not exhibited (726 = ? in BST integers)
6	c^2W/s^2W = 10/3 EXACT	EW mixing	exact	D-tier	rank·n_C/N_c = 10/3 (mechanism exhibited; this one is genuinely D-tier)

Per-entry assessment:

- Entry 6 (c^2W/s^2W = 10/3 EXACT, rank·n_C/N_c): mechanism cleanly exhibited (rank·n_C and N_c are BST primaries; ratio is forced). Stays D-tier.
- Entries 1-5: each has BST-primary-form decomposition + sub-percent or structural match, BUT mechanism chain to D_IV^5 is either partial (entries 1, 2, 4) or genuinely missing (entries 3, 5). Per Cal #20 already, entry 1 stepped down to I-tier. Per W3 criterion (mechanism chain to D_IV^5), entries 2-5 each need either:
  - (a) Mechanism chain exhibited at audit-defensible level → keep D-tier
  - (b) Step down to I-tier with "mechanism partial" label
  - (c) Step down to S-tier with "structural pattern without mechanism" label

Per Cal's three criteria (adapted to component-level audit)

Criterion 1 (Embedding): Each Δr entry must embed into D_IV^5 structure via a BST-primary mechanism chain. Currently: - Entry 6: SATISFIED (rank·n_C/N_c clean) - Entries 1-5: PARTIAL or MISSING

Criterion 2 (Mechanism): For each entry, mechanism must be a derivable closed-set operation on D_IV^5 producing the value. Currently: - Entries 1, 2, 4: mechanism partial (BST primary decomposition exists, but WHY this combination not exhibited) - Entries 3, 5: mechanism missing (1/107 and 1/726 don't decompose cleanly to BST primaries)

Criterion 3 (Forcing): BST primaries decompose uniquely (or as best-fit per Cal coincidence-filter Mode 2). Each entry checked: - Entries 1, 2, 4, 6: BST primary form is unique among small-integer candidates within precision tier - Entries 3, 5: 107 and 726 don't have clean BST primary forms — may be artifacts of the residual being a calculation difference, not a structural identification

K56 verdict per entry

Entry	Verdict	Rationale
1 α^{-1}	I-tier (already per Cal #20)	Mechanism partial; correction term not fully derived
2 $\Delta\alpha$	D → I-tier walk-back	Mechanism partial; substrate-coupling derivation needs Bergman line bundle anchor
3 $\Delta\rho = 1/107$	D → S-tier walk-back	107 has no BST primary decomposition; this is residual pattern, not structural identification
4 y_t = 136/137	D → I-tier walk-back	"1 − 1/N_max" form is BST primary, but WHY y_t specifically (vs y_b, y_c, etc.) needs mechanism — top Yukawa singled out structurally?
5 Δr_rem = -1/726	D → S-tier walk-back	726 has no BST primary decomposition; residual pattern, not structural identification
6 $c^2W/s^2W = 10/3$	D-tier stays	Mechanism cleanly exhibited via rank·n_C/N_c

Aggregate: 4 D→I/S walk-backs, 1 stays D-tier, 1 already I-tier per Cal #20.

The Cathedral component-level closure aggregate claim ("12/12 components close") survives, but at honest I-tier rather than D-tier when reported across the table. Specifically: - D-tier: 1 entry (c^2W/s^2W) + the structural Cathedral I-tier claim (12 components fitting the framework at varying precision tiers) - I-tier: 3 entries ($\Delta\alpha$, y_t, α^{-1} correction) at partial mechanism - S-tier: 2 entries ($\Delta\rho$, Δr_rem) at residual pattern without mechanism

Implications for Paper #115 v0.5+ and Cathedral Section 7

Lyra (Paper #115 v0.5 author): incorporate K56 verdict per-entry. Section 7 framing should change from "Cathedral 12/12 component-level closure at D-tier" to: - "Cathedral framework closes 12 components across precision tiers (1 D-tier, 3 I-tier, 2 S-tier residual, plus 6 prior-D-tier components from older Cathedral work; per-component tiers in Section 7.1 table). Aggregate Cathedral I-tier identification."

Cal's gate-pass on Paper #115 v0.5 was pending; K56 verdict is the per-entry calibration that lets it pass.

Elie (Toy 3012 author): Toy 3012 reports Δr table with current labels. Update toy output to reflect K56 per-entry tiers in next data refresh.

Grace (catalog): update INV catalog entries for the 5 walked-back items. D-count drops by 4 (from current state). I-count gains 3, S-count gains 2.

Per Casey's standard

- Simple: per-entry tier-check is mechanical (mechanism chain exhibited? Y/N).
- Works: catches the 4 walk-back cases cleanly; preserves the 1 genuine D-tier (Entry 6).
- Hard to break: tier-discipline criteria (Cal W1-W4 from K55) apply uniformly per-entry.
- Counter-example: none offered. (Entry 6's $c^2W/s^2W = 10/3$ is the counter-example of "mechanism exhibited" — confirms criterion fires correctly when it should.)

K56 verdict: WALK-BACK at component-level

4 entries walked back D→I ($\Delta\alpha$, y_t) and D→S ($\Delta\rho$, Δr_rem).

1 entry stays D-tier ($c^2W/s^2W = 10/3$, mechanism exhibited).

Aggregate Cathedral 12-component closure survives but as I-tier framework with per-component tier mix, not as D-tier aggregate.

Paper #115 v0.5 Section 7 framing updates per K56 verdict before Cal gate-pass.

This is consistent with Cal verdict #23 (muon g-2 D→I walk-back yesterday) and Lyra's read-pass on Paper #120 §4.4 (structural identification with mechanism-multi-week-open framing). Same discipline shape across the audit chain.

Action items

1. Lyra: update Paper #115 v0.5 Section 7 framing per K56 (4 walk-backs + 1 stay + aggregate I-tier label). When done, Cal gate-pass becomes available.
2. Elie: refresh Toy 3012 output to reflect K56 per-entry tiers.

3. Grace: catalog hygiene — update 5 INV entries with new tier labels; recompute D-tier percentage.
4. Keeper (me): file K57 next (Bridge Object tier definition audit).
5. Cal: standing — when Paper #115 v0.5 is updated, gate-pass + ratify K56 verdict.

K56 status

Walk-back ruling filed. 4 D→I/S downgrades, 1 D-tier stays, aggregate Cathedral I-tier framework preserved. Paper #115 v0.5 update required before external-survivability gate-pass.

— Keeper, 2026-05-19 Tuesday 10:40 EDT